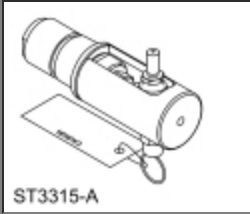
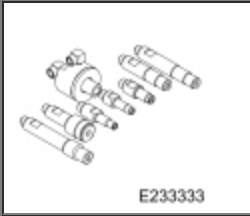


Torque Converter Contamination Inspection

 [Ford Ecat Electronic Parts Catalog](#)

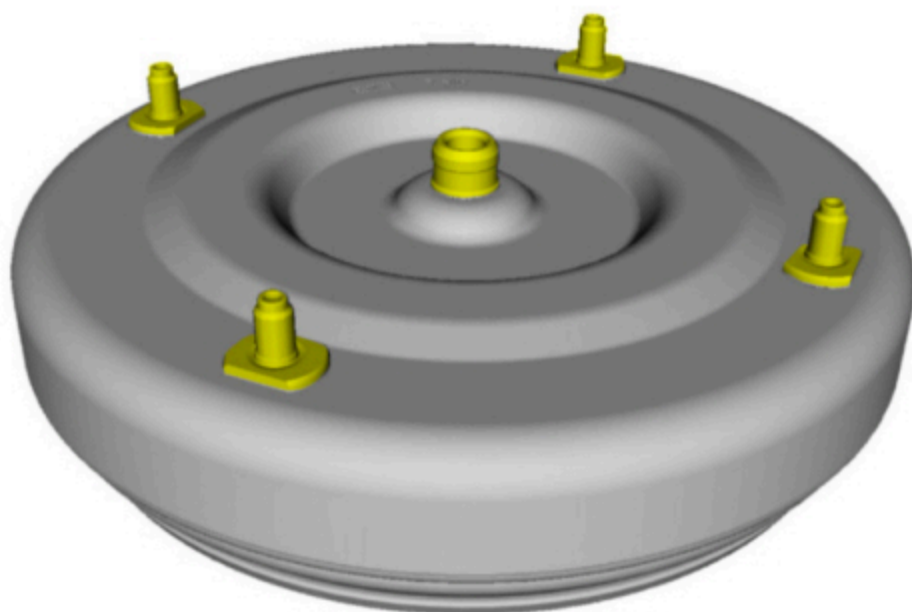
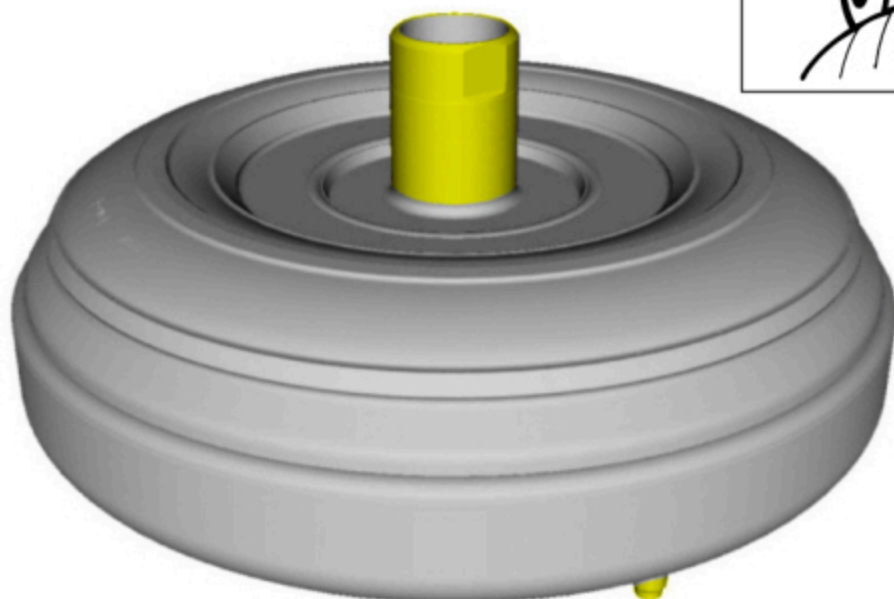
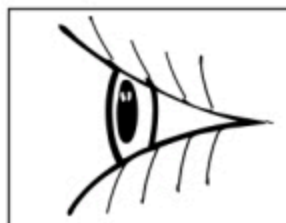
Special Tool(s)

 ST3315-A	Leak Tester, Torque Converter 307-691
 E233333	Tool Kit, Torque Converter Flusher 307-732

Material

Item	Specification
Motorcraft® MERCON® LV Automatic Transmission Fluid XT-10-QLVC (US); CXT-10-LV12 (Canada)	MERCON® LV

1. A new or remanufactured torque converter must be installed if one or more of the following statements is true:
- The sealing surface has a groove worn from the seal.
 - A torque converter malfunction has been determined based on complete diagnostic procedures.
 - The torque converter stud or studs, threaded pads, impeller hub or bushing are damaged.
 - The torque converter exhibits external discoloration (due to overheating).
 - There is evidence of water or antifreeze contamination.



E150775

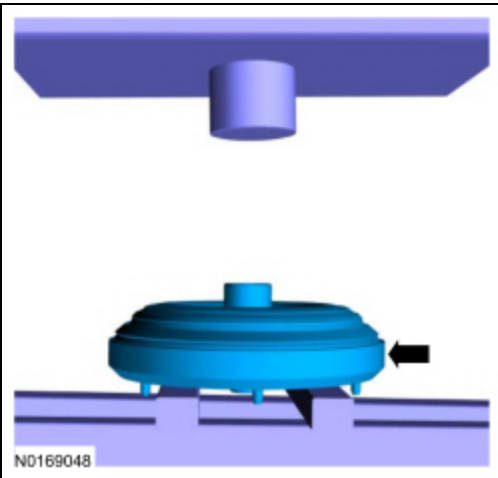
Flush The Torque Converter With The Transmission Cooling System Heated Flusher

2. NOTE: Use transmission fluid specified for this transmission. Do not use any supplemental transmission fluid additives or cleaning agents. The use of these products could cause internal transmission components to fail, which will affect the operation of the transmission.

The torque converter must be flushed every time the transmission is overhauled. It is mandatory that proper equipment and procedures be followed when flushing the torque converter. The flushing equipment used **MUST**:

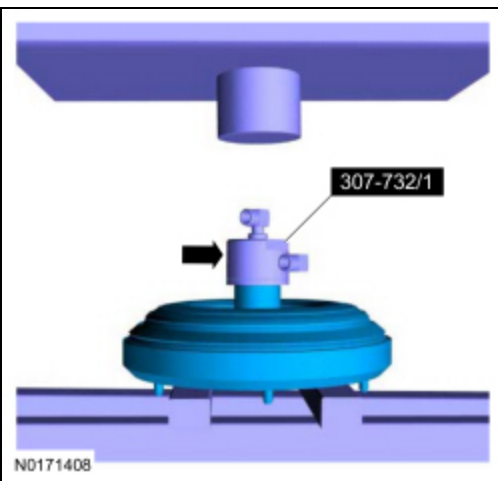
- Maintain the transmission fluid at 140°F or above
- Pulsate the transmission fluid during cleaning
- Have a GPM flow meter
- Have a filter with a rating of 100 micron or less
- Have air purge capability before and after flushing

3. If equipment meeting the specifications above is not available, the torque converter must be flushed by hand. Go to Flush The Torque Converter By Hand steps later in this procedure.
4. Check and top off the transmission fluid level of the transmission cooling system heated flusher with transmission fluid.
5. Turn on the heater and allow the transmission fluid in the transmission cooling system heated flusher 15-30 minutes to heat up to 60°C (140°F) before using.
6. Place the torque converter in an arbor press. Support the torque converter on the mounting pads.



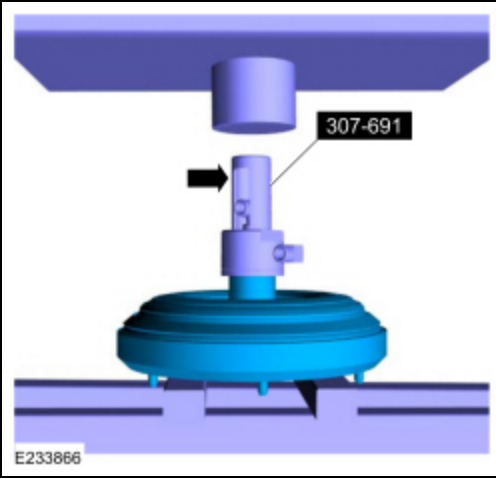
7. Using the special tools, assemble the correct turbine shaft simulator to the torque converter flush main hub and place it on the torque converter hub.

Special Tool(s): Tool Kit, Torque Converter Flusher 307-732

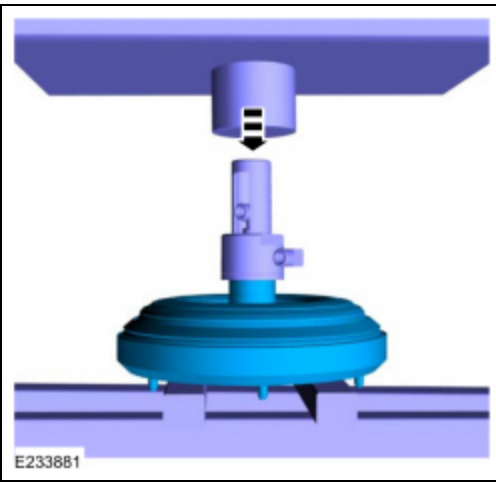


8. Using the special tool, install the slotted cap.

Special Tool(s): Tool Kit, Leak Tester, Torque Converter 307-691

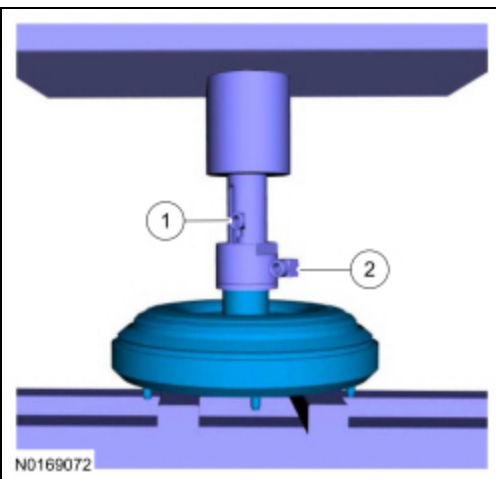


9. Apply enough force from the press to seal the torque converter flush main hub to the torque converter hub.



10. Connect the flush machine to the main hub.

1. Red hose on top.
2. Blue hose on bottom.



11. Follow the equipment instructions to purge the transmission fluid from the torque converter prior to starting the flushing procedure.

12. **WARNING: The torque converter, adaptor 307-732, and the hoses will be hot.**

NOTE: Maintain visual contact with torque converter during the entire flush procedure. Immediately stop the flush machine if a leak develops. Repeat set up steps to reseal the tool to the converter hub and continue flushing.

Forward flush the converter for 15 minutes.

13. Monitor GPM flow meter periodically during the flush procedure. Flow rate above 2.0 gallons per minute is required to break up and dislodge any contamination trapped behind the TCC (torque converter clutch) plate. Service flush machine filter(s) if flow rate drops below 2 GPM.

14. Follow the equipment instructions to purge the torque converter.

15. **WARNING: The torque converter, adaptor 307-732, and the hoses will be hot.**

Allow torque converter and equipment to cool for 30 minutes before handling.

16. Disconnect the hoses and remove the special tools.

Flush The Torque Converter By Hand

NOTICE: Do not use water-based cleaners or mineral spirits to clean or flush the torque converter or transmission damage will occur. Use only clean transmission fluid designated for the transmission and torque converter being serviced.

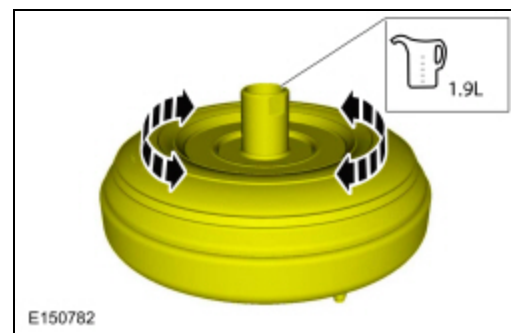
NOTE: Only flush the torque converter by hand when the transmission cooling system heated flusher is not available.

17. Pour a small amount of transmission fluid from the torque converter onto an absorbent white tissue or through a paper filter.

18. Examine the transmission fluid for contaminants. The transmission fluid must be free of metallic contaminants. If metallic contaminants are present, do not continue with hand flushing. The torque converter must be flushed with the transmission cooling system heated flusher.

19. Drain the remaining transmission fluid from the torque converter.

20. Using only the recommended transmission fluid, add 1.9L (2 qt) of clean transmission fluid into the converter and agitate by hand.



21. Thoroughly drain the transmission fluid.